

Alexander Ricciardi

Computer science, software development, and applied AI/ML professional

PYTHON | JAVA | C++ | AI/ML | AI AGENTS | RAG

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Profile

Computer science professional and a Master of Science student in Artificial Intelligence and Machine Learning. Has a Bachelor of Science in Computer Science, Summa Cum Laude, and an Associate of Science in Computer Science, High Distinction. Technical field-service experience since 2008. Seeking opportunities in software engineering and applied AI, with particular interest in AI agents, retrieval-augmented generation (RAG), vector search, LLM application development, and model integration with providers such as OpenAI, Anthropic, and DeepSeek. Also interested in data and business intelligence, research, internships, technical training, bootcamp instruction, and adjunct teaching while completing the master's degree.

Professional interests and subject areas

Software engineering and development; AI engineering; AI agents and agentic systems; retrieval-augmented generation (RAG); vector search, VectorRAG, GraphRAG, and knowledge graphs; LLM application development and model/API integration with OpenAI, Anthropic, DeepSeek, and other providers; artificial intelligence and machine learning; data analysis and business intelligence; AI research; Python, Java, C++, SQL, databases, data structures and algorithms; technical instruction, adjunct teaching, bootcamp instruction, employee AI training, generative AI and chatbot use, and responsible technology use.

Education

Master of Science in Artificial Intelligence and Machine Learning Colorado State University Global	Expected Graduation March 14, 2027
Bachelor of Science in Computer Science Colorado State University Global Summa Cum Laude	August 2025
Coursework. Operating Systems and Architecture; Software Engineering; Database Management; Networks; Cybersecurity; Java; Python; C++; Artificial Intelligence; Data Structures and Algorithms.	
Associate of Science in Computer Science Laramie County Community College High Distinction Honors	December 2023
Certificate of Completion, Full Stack Web Development The Array School of Technology and Design	June, 2018

Academic projects

CubeSat telemetry anomaly detection CSC510 course AI portfolio project, Colorado State University Global	2026 - current
• Developing a Python decision-support program that uses neural-network classification and A* search to analyze CubeSat telemetry anomalies.	
Mining Regulatory Compliance Assistant Bachelor of Science capstone project	2025
• Built a mine regulatory AI-powered assistant that queries Mine Safety and Health Administration regulations.	
• Combined an Advanced Parallel HybridRAG – Intelligent System that combines VectorRAG and GraphRAG retrieval with context fusion using Python, FastAPI, Neo4j, Streamlit, and Docker.	
Public computer science academic portfolio GitHub portfolio repo. showcasing code, papers, and projects.	2022 - current
• Publicly provides coursework and projects in Python, Java, C++, artificial intelligence, algorithms, operating systems, databases, networks, software engineering, cybersecurity, Android development, and graphics.	

Professional experience

Copier Technician Rocky Mountain Business Equipment Technical field service since 2008	2008 - present
• Repair, perform preventive maintenance on, and install office equipment for business customers.	
• Service Kyocera copiers, printers, facsimiles, scanners, and related multifunction systems.	

Master's Progress

CSC510 Foundations of Artificial Intelligence	In progress
CSC507 Foundations of Operating Systems	A / 4.0
CSC506 Design and Analysis of Algorithms	A / 4.0
CSC505 Principles of Software Development	A / 4.0
CSC502 Ethical Leadership in Software Development	A / 4.0
CSC501 Management for the Computer Science Professional	A / 4.0
CSC500 Principles of Programming, Python	A / 4.0

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Skills

Programming. Python; Java; C++; Kotlin; JavaScript; SQL; HTML; CSS.

AI and data. TensorFlow; PyTorch; NumPy; Pandas; Jupyter; Matplotlib; Plotly; neural networks; retrieval-augmented generation (RAG); VectorRAG; GraphRAG; vector search; knowledge graphs; AI agents; LLM application development.

Development. FastAPI; Streamlit; Docker; Neo4j; MySQL; Git; GitHub.

Academic communication. Technical reports; APA research writing; public code examples; project documentation.

Links

Omegapy GitHub. github.com/Omegapy - Software, AI, and academic projects.

Academic portfolio. github.com/Omegapy/My-Academics-Portfolio - Coursework, source code, papers, and project records.

AngryOwlAI. github.com/AngryOwlAI - AI project organization.

Code Chronicles. alexomegapy.com - Personal technical blog.

Medium. medium.com/@alex.omegapy - Technical articles and commentary.

Languages

English, French, and Spanish: fluent. Italian: spoken.